# STA304 Midterm Test (Lec0101)

Tuesday October 18, 2022

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*Instructions:*

- Total points: 20
- The test is 75 minutes long.
- There are 12 pages including this page.
- Questions begin on the next page.
- Only this exam book will be marked. Write your answers only in the space provided in this booklet. Any work on the Formula Sheet booklet will not be graded.
- Total questions: 12 (10 multiple choice, 2 short answer)
- Please ensure you **fill in your multiple Choice selections (Q1 - Q10) in the scantron on page 12**. Multiple choice answers NOT on the scantron will NOT be graded.
- Please use a calculator and do not round numbers in calculations until the very end, to avoid rounding error.
- In the multiple choice options, please select the closest answer (if the options are numeric) or the most appropriate answer (if the options are text).
- If you believe there to be a typo please still answer the question (see bullet point above), but you can list the question here and we will look into it after:

## Multiple Choice Questions

### Q1 (1 point)

What is the sample size required to be 99% confident of the population mean, within 4 units, if we are using systematic sampling on a population of size 600 with population **standard deviation** of 5?

a. 3
b. 6
c. 11
d. 24
e. 39

### Q2 (1 point)

The following question was asked of a random sample of University of Toronto students: "100% of Rotman graduates are employed within one month of graduation. Are you interested in pursuing studies at Rotman? Yes or No".

Which of the following statements are true?

a. The question is a leading question because it is asking multiple questions.
b. The question is a loaded question because it is asking multiple questions.
c. The question is a leading question because it pressures the reader to answer "Yes".
d. The question is a loaded question because it pressures the reader to answer "Yes".
e. The question is neither a leading question nor a loaded question.

### Q3 (1 point)

Assume a Simple Random Sample (without replacement) was conducted on kindergarten children. Assume the study variable of interest is age and the population size was N=10 The collected sample was: {4, 4, 5, 3}.

Calculate the lower bound of the 90% confidence interval for the overall population mean.

a. 3.32
b. 3.48
c. 3.52
d. 3.67
e. 4.52

## Q4 (1 point)

A company has surveyed its employees by stratifying by whether the employee is a manager or not. The company has 1000 employees in total, 50 of which are managers. The company is interested in estimating the mean salary of its employees. Which ONE of the following lines of code is most appropriate to provide the summaries needed for the different strata?

a. The code chunk below:

```
company_data %>%
  select(manager) %>%
  mutate(n=n(), y=mean(salary), s=sd(salary))
```

b. The code chunk below:

```
company_data %>%
  filter(manager=="Yes"|manager=="No") %>%
  summarize(n=n(manager))
```

c. The code chunk below:

```
company_data %>%
  filter(manager=="Yes"|manager=="No") %>%
  select(length(salary), mean(salary), sd(salary))
```

d. The code chunk below:

```
company_data %>%
  select(salary) %>%
  group_by(manager) %>%
  summarize(n=n(), y=mean(salary), s=sd(salary))
```

e. The code chunk below:

```
company_data %>%
  group_by(manager) %>%
  summarize(n=n(), y=mean(salary), s=sd(salary))
```

## Q5 (1 point)

A company has surveyed its employees while stratifying by whether the employee is a manager or not. The company has 1000 employees in total, 50 of which are managers. The company is interested in estimating the mean salary of its employees. Use the following summary table of the salary variable (from the previous question) to calculate the upper bound of the 95% CI for the overall mean salary.

| manager | n | mean | sd |
|---|---|---|---|
| Yes | 5 | 93.1 | 2.4 |
| No | 95 | 56.8 | 5.6 |

a. 57.595
b. 58.647
c. 58.937
d. 59.047
e. 59.635

## Q6 (1 point)

Which ONE of the following are TRUE?

a. Snowball sampling is a non-probability based sampling approach that is helpful when trying to survey underground populations, such as the homeless.
b. Convenience sampling is a probability based sampling approach that is helpful when trying to meet quotas in the population.
c. In cluster sampling, we want to ensure that we are selecting some observations from each cluster.
d. Quota sampling is a non-probability based sampling approach that randomly selects observations from different strata in the population.
e. It is not a good idea to test your survey before you implement it to your sample.

## Q7 (1 point)

A study wishes to run a hypothesis test to make inference on the mass to height ratio, based on a simple random sample of $n = 68$, from a population is of size $N = 5000$. Which ONE of the following lines of code is most appropriate at generating the ratio estimate and standard error of interest?

a. The code chunk below:

```
fpc.srs <- rep(5000/68)
my.design <- svydesign(id=~1, data=my_data, fpc=fpc.srs)
svyratio(~mass, ~height, design=my.design)
```

b. The code chunk below:

```
my.design <- svydesign(id=~1, data=my_data)
svyratio(~mass, ~height, design=my.design)
```

c. The code chunk below:

```
my.design <- svydesign(id=~1,
                       strata=~strata,
                       weights = ~samp_strata_weight,
                       data=my_data,
                       fpc=~pop_strata_weight)
svyratio(~mass, ~height, design=my.design)
```

d. The code chunk below:

```
my.design <- svydesign(id=~1, data=my_data, fpc=rep(5000, 68))
svyratio(~mass, ~height, design=my.design)
```

e. The code chunk below:

```
lm(~mass, ~height)
```

## Q8 (1 point)

A company of $N = 20$ employees wishes to take a random sample of 3 different employees. To do this they put every employees' name on a piece of paper in a hat and then shuffle the names and select 3 pieces of paper all at once (i.e., without returning the names to the hat after selection). What is the probability of any one sample of 3 names is selected?

a. 0.00013
b. 0.00088
c. 0.00338
d. 0.03704
e. 0.05000

## Q9 (1 point)

Which ONE of the following models, best describes what the code below is estimating?

Note the data has the following variables:

- `passed_course` is a categorical variable with 2 categories (1 if they passed, and 0 if not).
- `program_of_study` is a categorical variable with 3 categories (Major, Minor, Specialist).
- `hours_slept` is a numeric variable.

```
my_model <- glm(passed_course ~ hours_slept + program_of_study, family="binomial")
summary(my_model)
```

a. $log(\frac{p}{1-p}) = \beta_0 + \beta_1 x_{hours} + \beta_2 x_{minor} + \beta_3 x_{specialist}$
b. $log(\frac{p}{1-p}) = \beta_0 + \beta_1 x_{hours} + \beta_2 x_{minor} + \beta_3 x_{specialist} + \epsilon$
c. $y = \beta_0 + \beta_1 x_{hours} + \beta_2 x_{minor} + \beta_3 x_{specialist} + \epsilon$
d. $log(\frac{p}{1-p}) = \beta_0 + \beta_1 x_{hours} + \beta_2 x_{major} + \beta_3 x_{minor} + \beta_4 x_{specicialist} + \epsilon$
e. $y = \beta_0 + \beta_1 x_{hours}$

**Q10 (1 point)**

Which ONE of the following best describes the difference between a Bayesian and Frequentist approach?

a. A Frequentist approach invokes a prior distribution on the parameter of interest and then updates the distribution based on collected data.
b. A Bayesian approach invokes a prior distribution on the parameter of interest and then updates the distribution based on collected data.
c. There is no difference
d. It is always best to use a Frequentist approach.
e. It is always best to use a Bayesian approach.

## Short Answer Question

### Q11 (5 points total)

A company is interested in learning about it's employees' commute times to work. The company has 80 employees in 4 different departments (all of equal size) and can only afford to survey 40 employees.

a. (2 points) The company is not sure if they should treat departments as clusters or strata. Explain in 2 or 3 sentences what advice would you give the company to help them decide between stratified and cluster sampling.

Note: Please ensure you demonstrate your understanding of the difference between stratified and cluster sampling.

## Short Answer Question

### Q11 (5 points total)

A company is interested in learning about it's employees' commute times to work. The company has 80 employees in 4 different departments (all of equal size) and can only afford to survey 40 employees.

b. (3 points) If the company decided to treat departments as *clusters* describe how they could use the `sample()` function to select their clusters. Note: if you feel more comfortable you could write a line of code here and explain it.

## Short Answer Question

### Q12 (5 points total)

Assume you have a sample of $y_1, y_2, ..., y_n$ taken from the population of $\{u_1, u_2, ..., u_N\}$. Recall, $E(Y_i) = \mu$, $Var(Y_i) = \sigma^2$.

a. (1 point) Show that $E(Y_i) = \mu$. I.e., Show that the expected value, of a sample of size $n = 1$, is the population mean.

**Q12 (5 points total)**

Assume you have a sample of $y_1, y_2, ..., y_n$ taken from the population of $\{u_1, u_2, ..., u_N\}$. Recall, $E(Y_i) = \mu$, $Var(Y_i) = \sigma^2$.

b. (4 points) Derive $E(3Y_i^2)$. Write it as an expression of $\mu$ and $\sigma^2$.

*Recall:* $Var(Y_i) = E[(Y_i - E(Y_i))^2]$